Single Line Diagram (SLD)

Visual user manual - build the network, check calculations, and create drawing sheets



01 Build the panel hierarchy

02 Add and edit feeders

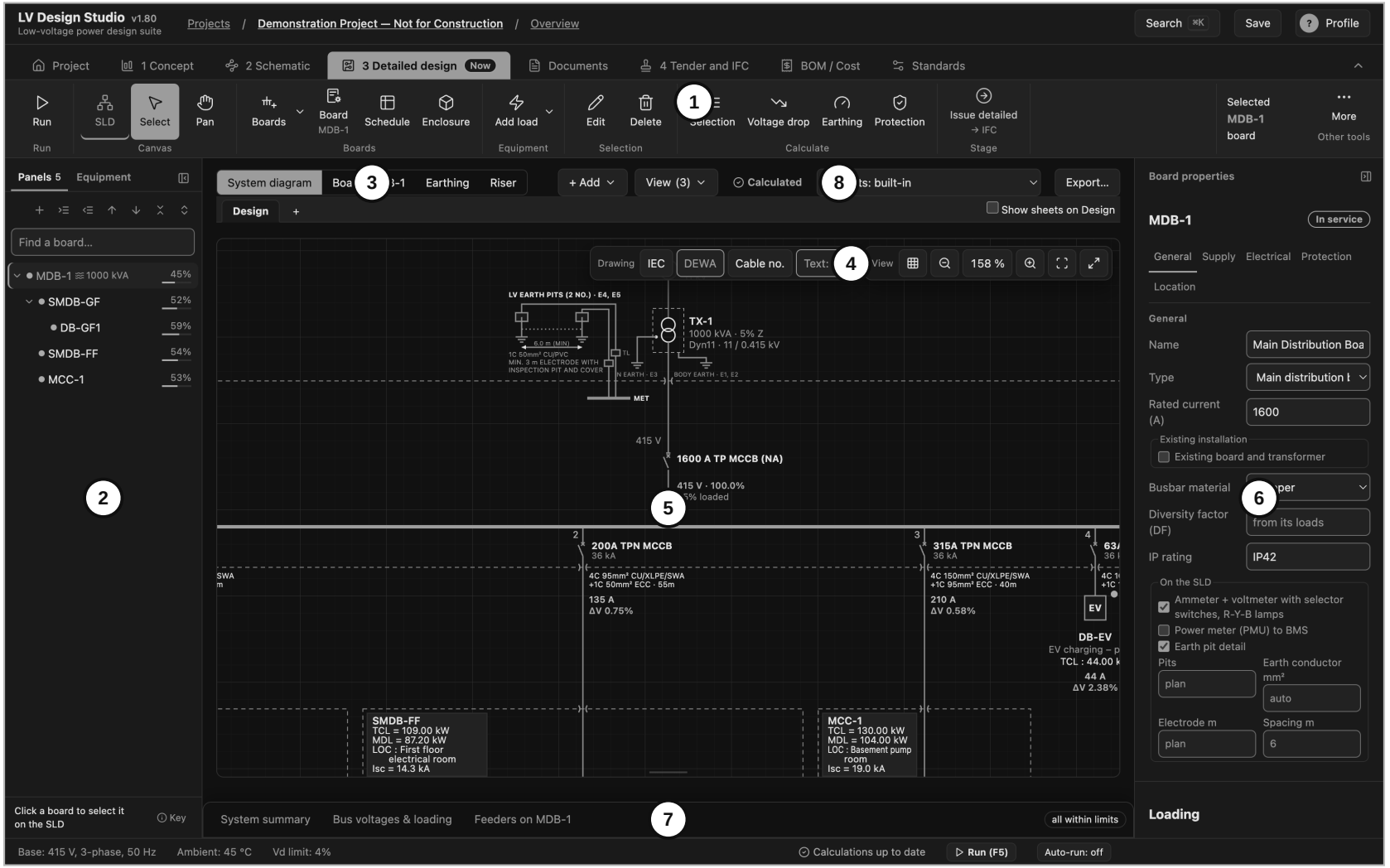
03 Run and review checks

04 Create and export sheets

START HERE

Know the SLD work area

The numbered markers identify the controls used to build and check the network.

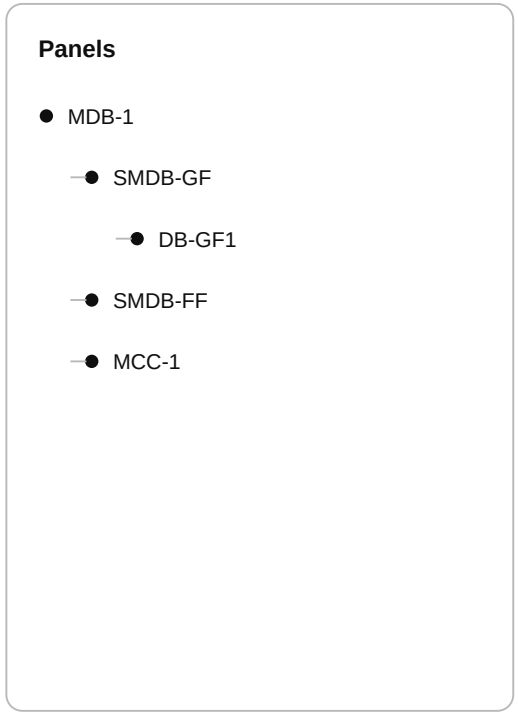


1. SLD and selection controls
2. Panel tree
3. System, board, earthing, and riser views
4. Drawing display tools
5. Interactive SLD canvas
6. Board properties or circuit results
7. System, bus, and feeder results
8. Export and drawing sheets

STEPS 1-3

Create the supply hierarchy before adding final loads

Start at the main board and work downstream. Every board needs an incomer feeder.



1 Select the parent board

Choose the board that will supply the new board. The selected board is highlighted in the panel tree and on the SLD.

2 Add the downstream board

Select Boards. Enter board ID, name, type, rating, source, incomer ID, cable, and breaker details.

3 Repeat or build in batches

Add boards one by one, or use Build hierarchy when creating several panels and connections together.

4 Check the supply chain

Open the board Supply tab. Confirm transformer, parent panels, incomer cable, fault level, and cumulative voltage drop.

Move an existing board

Drag a downstream board onto its new parent in the panel tree. Confirm the move, then check the incomer cable length.

STEPS 4-6

Add equipment, then complete cable and protection data

The feeder dialog gives a live summary of current, cable, breaker, and total voltage drop.

General & load

- Unique circuit ID and name
- Load type and connected kW
- Demand factor and power factor
- Essential, existing, or generation

Cable

- Length and cores
- Cable size and parallel runs
- Installation and cable type
- Protective conductor
- Suggest cable size

Protection

- Breaker rating, Icu, and type
- Instantaneous setting
- Circuit purpose
- RCD/ELCB and isolator

Metering

- Direct or CT-operated kWh meter

Fast editing on the SLD

Click to select. Double-click or press F2 to edit. Drag a board, load, or incomer to reconnect it. Delete removes the selected item. Ctrl/Command+C and Ctrl/Command+V copy and paste a board branch.

STEPS 7-8

Run calculations after every design change

Live totals update immediately, but network study results can become stale until F5 is run.



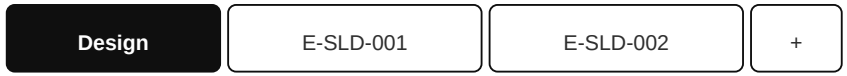
Checks to review

- Design current <= breaker <= cable capacity
- Total voltage drop within the project limit
- Breaker Icu above the calculated fault level
- Earth-fault loop and disconnection time acceptable
- Board loading and phase loading acceptable
- No stale-results notice before export

FINISH

Turn the working SLD into controlled drawing sheets

Design is the full editable canvas. Sheet tabs are the pages prepared for issue.



1 Select a panel

Choose the branch for the new sheet.

2 Add a sheet

Use +, then Selected panel and its branch, Empty sheet, or Split all panels.

3 Set controls

Check paper size, title, panels, cable labels, result layers, status, and revision.

4 Preview

Use Reports > Drawings to review every sheet before issue.

Quick SLD export

- Open the system diagram
- Run F5 if result labels are shown
- Select Export
- Complete title-block details
- Export SVG, DXF, or PDF

Drawing-set export

- Open Reports > Drawings
- Preview sheets
- Export one combined PDF
- Or export PDF/DXF per sheet
- Optionally include the drawing register